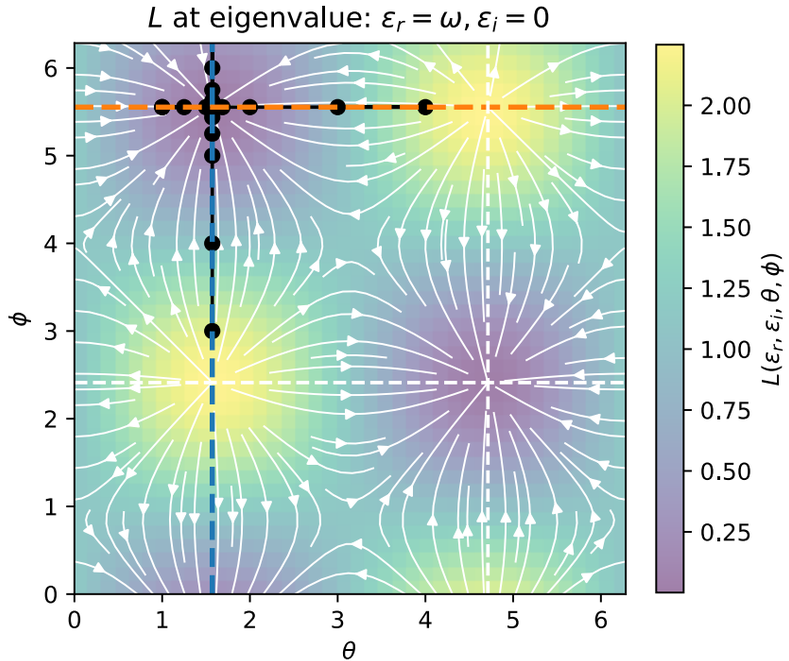
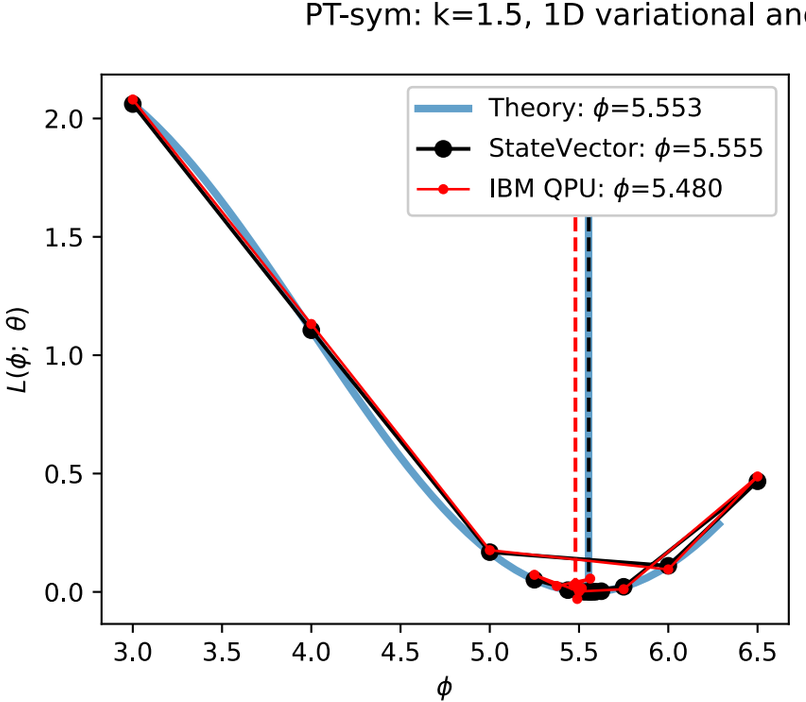
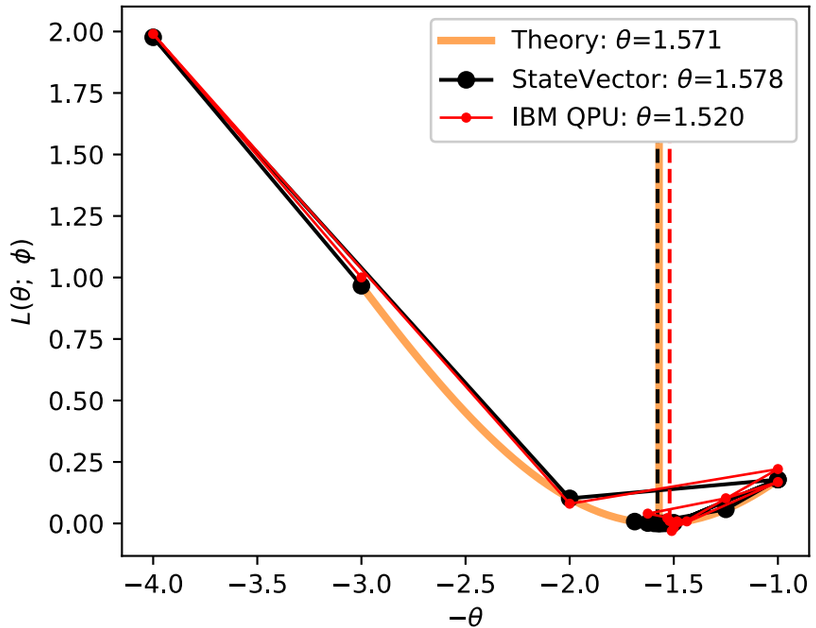

L at eigenvalue: $\varepsilon_r = \omega, \varepsilon_i = 0$
ϕ
θ
$L(\varepsilon_r, \varepsilon_i, \theta, \phi)$


PT-sym: k=1.5, 1D variational angle optimization, COBYLA 15 iterations
Theory: ϕ=5.553
StateVector: ϕ=5.555
IBM QPU: ϕ=5.480
$L(\phi;\ \theta)$
ϕ


Theory: θ=1.571
StateVector: θ=1.578
IBM QPU: θ=1.520
$L(\theta;\ \phi)$
$-\theta$